

Deadlock Detection.

```
#include <stdio.h>
#define MAX 10

int main() {
    int n, m;
    int allocation[MAX][MAX], request[MAX][MAX], available[MAX];
    int work[MAX], finish[MAX];
    int i, j, k;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter number of resource types: ");
    scanf("%d", &m);
    printf("Enter Allocation Matrix:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < m; j++) {
            scanf("%d", &allocation[i][j]);
        }
    }
    printf("Enter Request Matrix:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < m; j++) {
            scanf("%d", &request[i][j]);
        }
    }
    printf("\nEnter Available Resources:\n");
    for (j = 0; j < m; j++) {
        scanf("%d", &available[j]);
    }
}
```

```

for(j=0; j<m; j++) {
    work[j] = available[j];
}
for(i=0; i<n; i++) {
    finish[i] = 0;
}

int found;
do {
    found = 0;
    for(i=0; i<n; i++) {
        if(finish[i] == 0) {
            int canExecute = 1;

            for(j=0; j<m; j++) {
                if(request[i][j] > work[j]) {
                    canExecute = 0;
                    break;
                }
            }

            if(canExecute) {
                for(k=0; k<m; k++) {
                    work[k] += allocation[i][k];
                }
                finish[i] = 1;
                found = 1;
            }
        }
    }
} while(found);

```



```

int deadlock = 0;
printf("In Deadlock Processes:");
for(i=0; i<n; i++) {
    if(finish[i] == 0) {
        printf("P%d", i);
        deadlock = 1;
    }
}

if(deadlock == 0)
    printf("None (No deadlock)\n");
else
    printf("In System is in deadlock\n");
return 0;
}

```

output:

Enter number of processes: 4

Enter number of resource types: 3

Enter allocation matrix:

1 0 2

2 1 1

1 0 3

1 2 2

Enter request matrix:

0 0 1

1 0 2

0 0 0

3 3 0

Enter available resource:

0 0 0

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Deadlock Processes: P3

System is in deadlock.